SPECTRAL ESTIMATION

Spectrum analysis with a Kaiser window

Homework 01 | Direct window implementation, amplitude calibration, and signal reconstruction

Objective

Estimate the number of sinusoidal components Q, their amplitudes A_i, and their frequencies f_i from the supplied two-column timing waveform. The assignment requires a time-domain plot, spectra for at least four distinct Kaiser parameters, gain compensation, and Python source code [1].

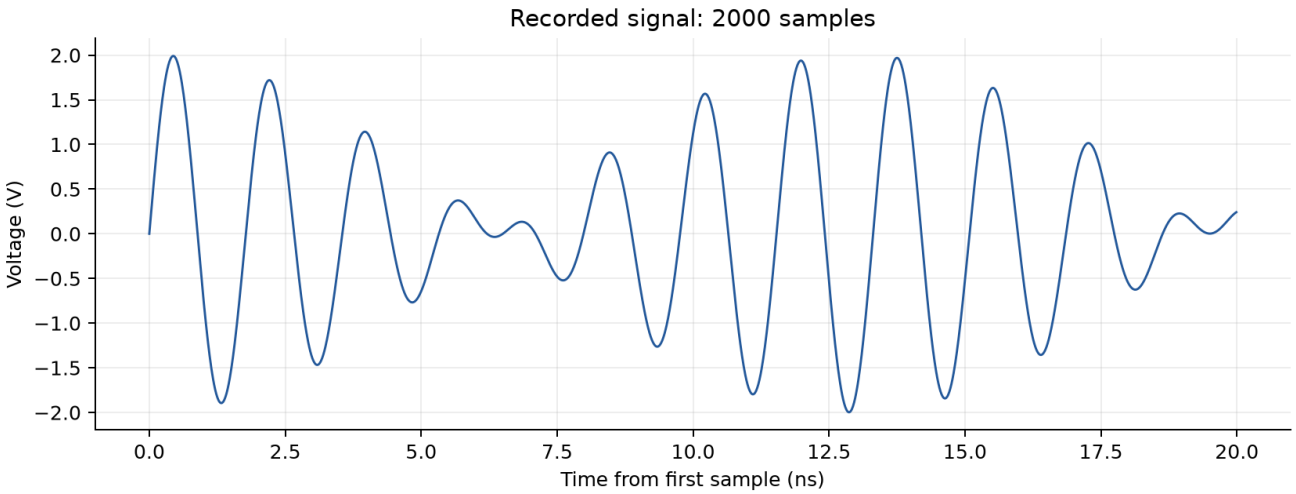


Figure 1. The supplied time-domain waveform. The varying envelope indicates multiple frequencies; the waveform alone does not determine their number or amplitudes.

Record property	Value	Interpretation
Samples	2,000	Uniformly spaced voltage observations
Sample interval / rate	10 ps / 100 GHz	Nyquist frequency: 50 GHz
Record length / endpoint span	20.00 ns / 19.99 ns	N Δt and (N - 1) Δt have different roles
Native DFT spacing	50 MHz	1 / (N Δt)

Result in brief: the combined window analysis and reconstruction support Q = 3. Two approximately 1 V components occur near 525 and 602 MHz, and a much weaker component occurs near 888.5 MHz. The final time-domain fit has RMSE 4.720e-15 V.

1. Windowing and amplitude calibration

Direct implementation

For sample indices n = 0, ..., N - 1, the implemented window follows the assignment's modified-Bessel equation directly. No built-in Kaiser-window constructor is used.

$$w[n] = I_0(\pi \alpha \sqrt{(1 - [2n/(N - 1) - 1]^2)}) / I_0(\pi \alpha)$$

This course uses α. Libraries that parameterize a Kaiser window by β use β = π α. Substituting α directly for β would create a different taper. The α = 0 case reduces to the rectangular window; increasing α reduces the endpoints and broadens the main lobe.

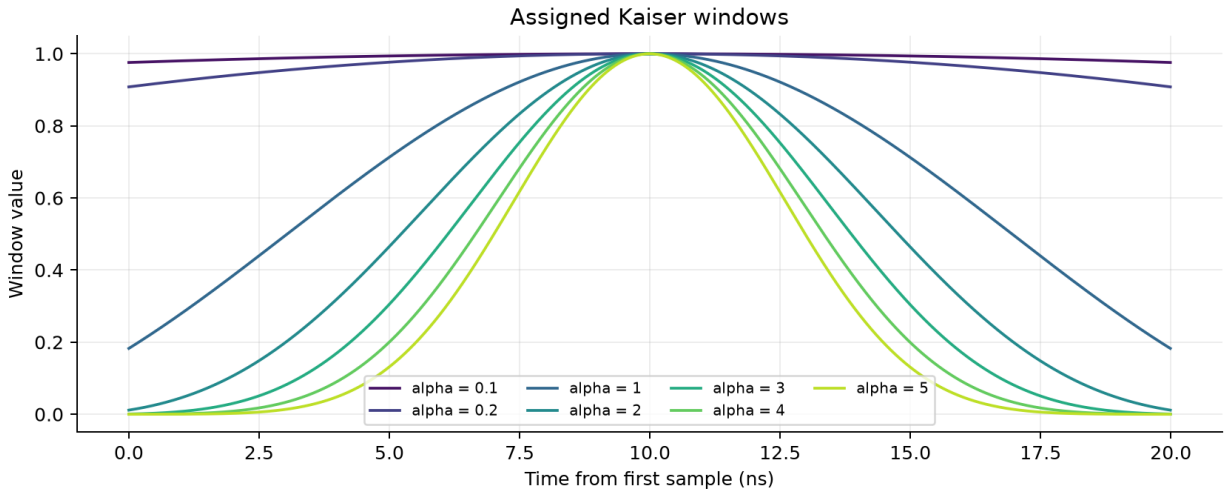


Figure 2. Directly evaluated Kaiser windows. The plotted parameter labels use the assignment's α convention.

Window gain and a one-sided spectrum

Let Y[k] be the M-point DFT of x[n]w[n], with zero padding when M > N. The coherent gain is $G_c = (1/N) \Sigma w[n]$. For an isolated, bin-centered sinusoid away from DC and Nyquist, a calibrated single-sided amplitude is

$$A[k] = 2 |Y[k]| / \Sigma w[n] = 2 |Y[k]| / (N G_c).$$

DC is not doubled. For even M, the Nyquist bin is not doubled either; an odd-length real FFT has no Nyquist bin. The denominator uses the N real samples and their window sum, never the padded length M. Zero padding contributes no additional signal samples.

The plotted vertical scale is $20 \log_{10}(A / 1\ V)$, labeled dBV. Thus 1 V corresponds to 0 dBV and 0.00316228 V corresponds to -50 dBV. The displayed floor is only a plotting limit; the fit uses the original voltage samples.

Calibration limits

Coherent-gain correction removes the taper's amplitude loss. It does not remove off-grid scalloping or bias from overlapping lobes. These effects motivate refinement with a time-domain sinusoidal model instead of reading every amplitude directly from a peak height.

2. The spectrum changes with α

The four individual spectrum figures on this page and the next use the same time record, frequency units, amplitude reference, and gain correction. Each figure labels its own α value, as required by the assignment.

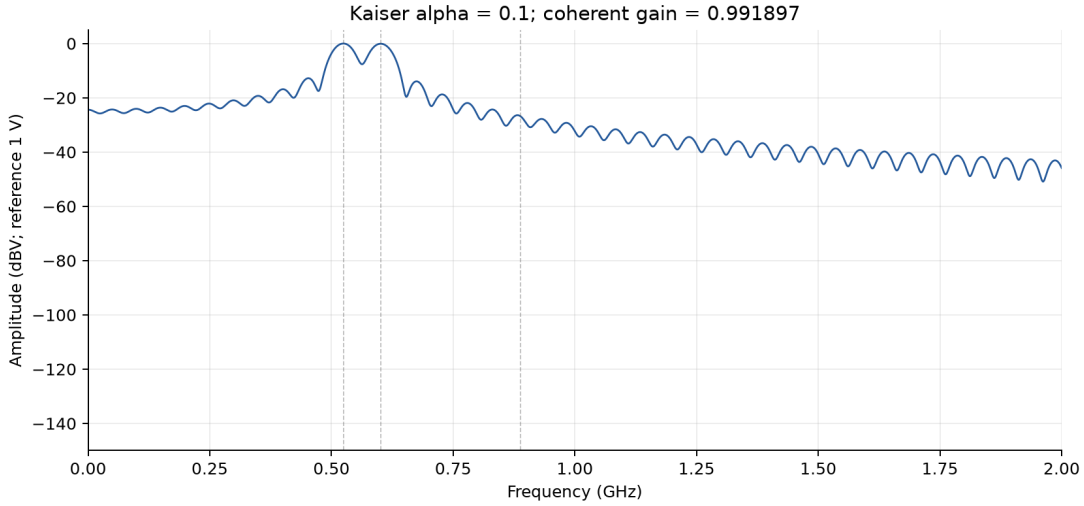


Figure 3. α = 0.1. A narrow main lobe helps distinguish the nearby strong pair, while sidelobes mask the distant weak component.

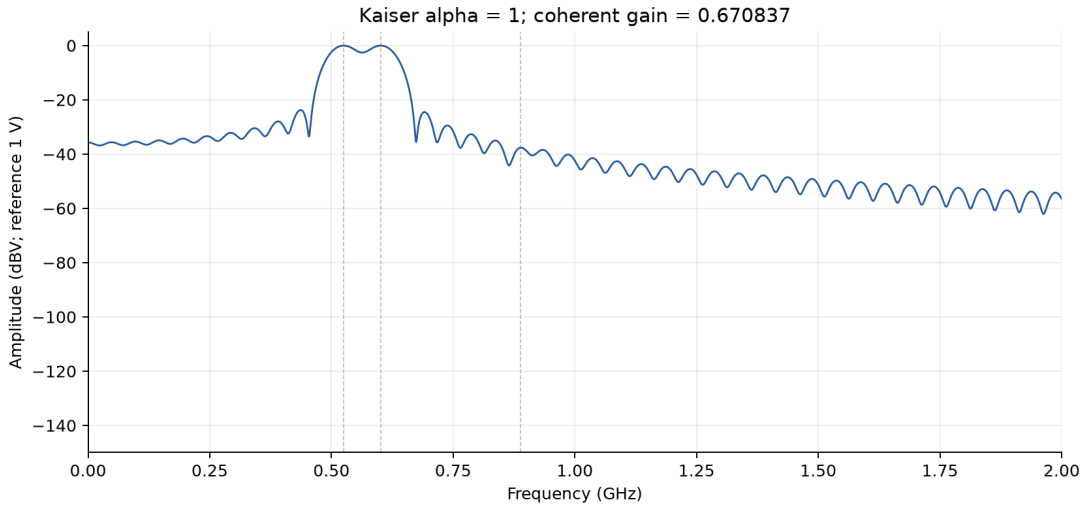


Figure 4. α = 1. Sidelobe levels decrease, with a wider main lobe than for α = 0.1.

Reading the comparison

Small α retains a narrow main lobe, which helps distinguish the two strong components separated by roughly 77 MHz. Large α lowers sidelobes so that leakage from the strong pair is less likely to mask a distant weak tone. The main lobe simultaneously widens, so no single window setting is best for both tasks.

Peak heights can change even after gain compensation when lobes overlap or a peak is sampled away from its maximum. The window comparison is therefore used to identify candidate structure; the final table comes from a joint fit.

3. Recovering the weak component

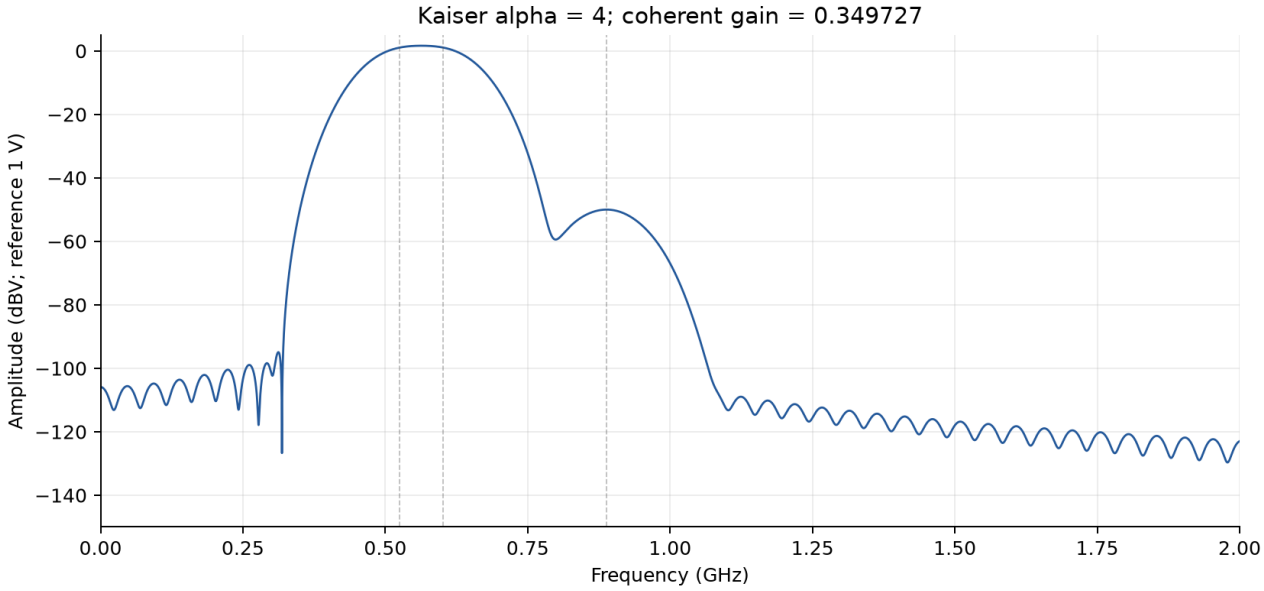


Figure 5. α = 4. Strong-tone leakage is reduced, making the weak feature near 0.8885 GHz visible on the calibrated amplitude scale.

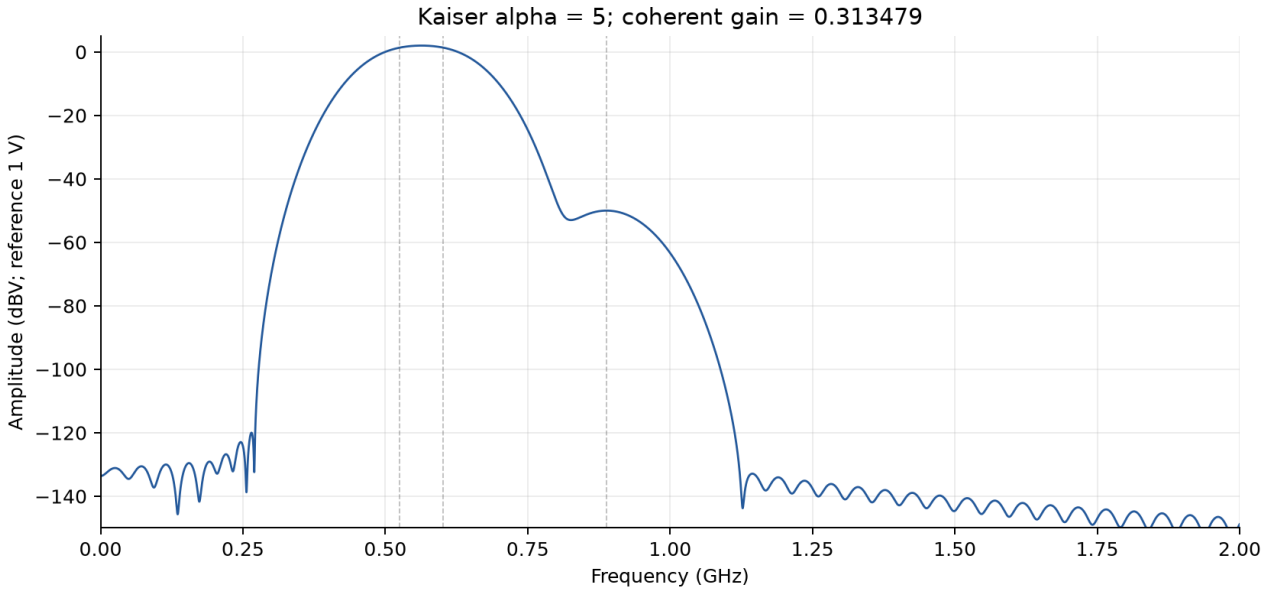


Figure 6. α = 5. The same weak feature persists under a different taper. This cross-check supports treating it as signal structure rather than relying on a single-window sidelobe.

Why Q = 2 is incomplete

A small-α spectrum can emphasize the two large peaks and obscure the third component. A model containing only the strong pair leaves a coherent residual; adding the weak component removes that residual and improves the numerical fit. Window comparison and residual verification are both needed to establish Q.

Zero padding and resolving power

The FFT uses M = 1,048,575 points, giving a displayed grid spacing of 0.0953675 MHz. The acquired record still lasts 20 ns and its native spacing remains 50 MHz. Padding interpolates the sampled spectrum; it does not narrow a window's main lobe or add independent information. The joint fit uses a parametric model and the full time record.

4. Estimated signal and reconstruction

Fitting procedure

Residual Kaiser spectral candidates; variable projection with bounded damped Gauss-Newton frequency refinement; BIC order selection

At each trial set of frequencies, the sine coefficients, cosine coefficients, and constant offset are solved by linear least squares. Frequency refinement is nonlinear. The fitted amplitude is $\sqrt{(s_i^2 + c_i^2)}$, and the phase is $\text{atan2}(c_i, s_i)$. This allows numerical checks of the zero-phase sinusoidal model in the prompt without forcing the phase in advance.

Component	Frequency (MHz)	Amplitude (V)	Amplitude (dBV)
1	525.000000	1.0000000000	0.0000
2	602.000000	1.0000000000	0.0000
3	888.500000	0.0031622777	-50.0000

Selected model: Q = 3. Fitted offset: -1.896e-16 V. Reconstruction RMSE: 4.720e-15 V. Maximum absolute residual: 2.737e-14 V. Relative RMSE: 4.781e-15.

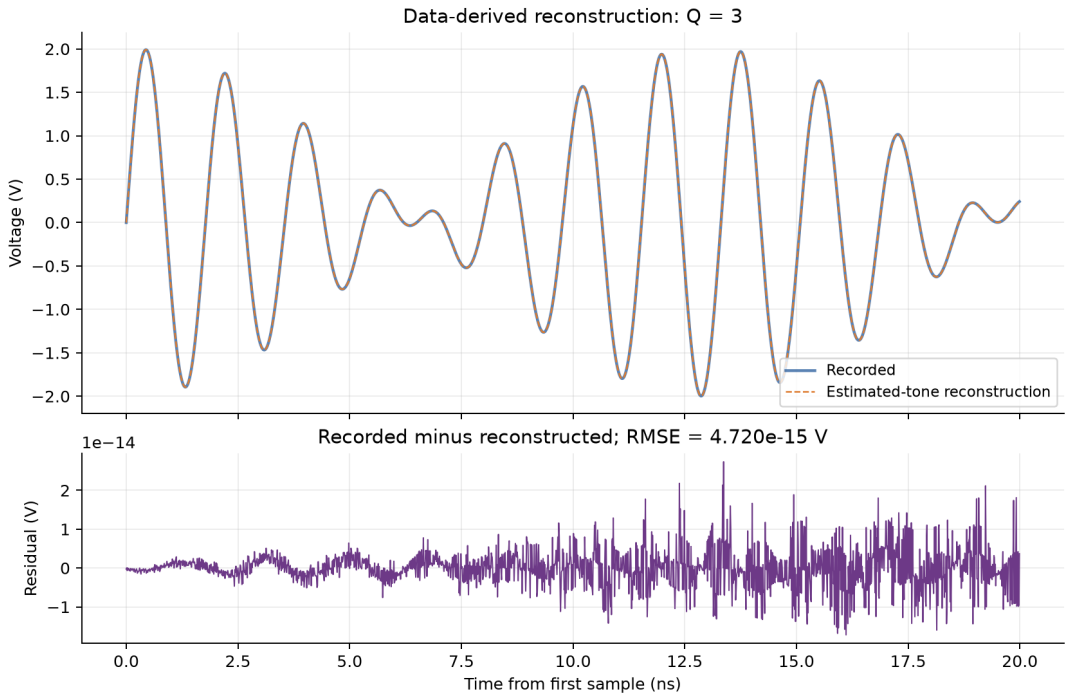


Figure 7. Reconstruction from all fitted components, together with the residual. The residual scale is reported explicitly so that an apparently perfect overlay is not mistaken for an independent accuracy check.

Precision and interpretation

The table retains enough digits to reproduce the numerical result. These are parameter estimates for a supplied deterministic waveform; a small residual does not establish corresponding measurement accuracy or uncertainty for a physical experiment. The fitted phases and model-order diagnostics are retained in the machine-readable results.

5. Reproducibility and verification

Numerical model-order evidence

Q	RMSE (V)	Residual sum of squares	BIC
0	9.867e-01	1.947e+03	5.667
1	6.887e-01	9.487e+02	-1409.461
2	2.228e-03	9.925e-03	-24322.285
3	4.720e-15	4.456e-26	-92027.395

The RSS column is in V^2. For BIC, the code uses unit-RMS-normalized samples and floors normalized RSS at $N \times (1e\text{-}10)^2$. It evaluates $N \ln(RSS_{norm,floored}/N) + (3Q + 1) \ln N$. The normalization changes the common scale, not the model-order comparison. The search stops when the relative numerical residual tolerance is reached; the physical-voltage RMSE remains unfloored.

Core amplitude-calibration algorithm

```
n = np.arange(N, dtype=float)
z = 2*n/(N - 1) - 1
w = np.i0(np.pi*alpha*np.sqrt(np.maximum(0, 1 - z*z)))
w /= np.i0(np.pi*alpha)
Y = np.fft.rfft(x*w, n=M)
A = np.abs(Y) / np.sum(w)
if M % 2 == 0:
    A[1:-1] *= 2      # DC and Nyquist stay undoubled
else:
    A[1:] *= 2        # no Nyquist bin for odd M
f = np.fft.rfftfreq(M, d=dt)
```

Repository evidence

The accompanying Python source performs input validation, computes every plot, estimates the model parameters, and writes summary.json and parameters.csv. Figures in this report are embedded directly from those generated result files. The repository README gives the command for reproducing the analysis and the checks; the complete source is supplied with the report, satisfying the code requirement.

Check	Evidence
Input integrity	N = 2000; uniformly increasing time; finite samples
Window definition	Direct modified-Bessel evaluation; β = π α
Amplitude handling	Coherent gain; correct DC/Nyquist treatment; volts and dBV
Signal completeness	Multiple α values plus joint-fit residual and model-order results

References

[1] Course-provided assignment, Time-frequency transform (1) - Using Kaiser window for spectral analysis. Source waveform: HW1_Kaiser.txt. The assignment and waveform are reference inputs; no personal identifiers are reproduced.